Optical-based thickness determination of MoO_3 nanosheets

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Abstract: Considering that two-dimensional (2D) molybdenum trioxide has acquired more attention in the last few years, it is relevant to speed up thickness identification of this material. We provide two fast and non-destructive methods to evaluate the thickness of MoO_3 flakes on SiO_2/Si substrates. First, by means of quantitative analysis of the apparent color of the flakes in optical microscopy images, one can make a first approximation of the thickness with an uncertainty of ±3 nm. The second method is based on the fit of optical contrast spectra, acquired with micro-reflectance measurements, to a Fresnel law-based model that provides an accurate measurement of the flake thickness with ±2 nm of uncertainty

Keywords: MoO_3, complex oxides, optoelectronics, 2D materials, layered materials

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1. Introduction

Since the isolation of graphene by mechanical exfoliation in 2004 [1] the catalogue of different 2D materials with complementary properties keeps growing [2–9]. Among them, wide bandgap semiconductor materials have attracted a great deal of attention due to their potential in optoelectronic applications [10] requiring electrically conductive materials that are transparent to visible light. Molybdenum trioxide (MoO_3) in its α-phase, is a van der Waals material with a monolayer thickness of ~0.7 nm [11,12] and a direct bandgap of approximately 3 eV [13–15], suitable for such applications [16]. It has been used, in thin films, to enhance the injection of holes in Organic Light-Emitting Diodes as buffer layer [17,18], in organic photovoltaics [19], perovskite solar cells [20] and silicon solar cells [21], besides it can be used in gas sensors [15,22,23]. Moreover, MoO_3 is interesting because of its photochromic, thermochromic, electrochromic effects [24–28], and catalytic properties in the partial oxidation of methanol to formaldehyde [29–33]. Sub-stoichiometric MoO_3 quantum dots have been synthesized as Surface Enhanced Raman Scattering substrates [34]. Furthermore, this material displays an in-plane anisotropy of the crystal structure in its layered phase (α-MoO_3) [16,35–38], which can be exploited to fabricate novel optical and optoelectronic devices [39–41], besides it has been observed anisotropic phonon polariton propagation along MoO_3 surface [42].

A rapid and non-destructive method to determine the thickness of MoO_3 would be highly desirable for the further development of this line of research. This is precisely the goal of this manuscript: to provide a guide to determine the thickness of MoO_3 nanosheets (in the 0-100 nm range) by optical microscopy-based methods. We propose two complementary approaches: first, a coarse thickness estimation based on the apparent interference color of the flakes, and, second a quantitative analysis of the reflection spectra using a Fresnel law-based model.

2. Materials and Methods

MoO_3 flakes were grown by a simple physical vapor transport method carried out at atmospheric conditions, developed in Reference [35]. A molybdenum foil is oxidized by heating it up on a hotplate at 540ºC, then a silicon wafer is placed on top. The molybdenum oxide sublimes and crystallizes on the surface of the Si wafer, at slightly lower temperature, forming MoO_3 flakes. These MoO_3 flakes can be easily lifted from the Si wafer surface with a Gel-Film (WF x4 6.0 mil, from Gel-Pak) viscoelastic substrate and subsequently transferred to and arbitrary target substrate. We have transferred the flakes onto silicon chips with 297 nm SiO_2 capping layer (see the Supporting Information for the quantitative determination of the SiO_2 thickness) as it is one of the standard substrates to work with 2D materials.

Atomic Force Microscopy

AFM characterization was performed at ambient conditions using two commercial AFM systems: (1) a Nanotec AFM system has been used [43] in dynamic mode with a NextTip (NT-SS-II) cantilever (resonance frequency of 75 kHz), (2) an ezAFM (by Nanomagnetics) AFM operated in dynamic mode with Tap190Al-G by Budget Sensors AFM cantilevers (force constant 48 Nm^{-1} and resonance frequency 190 kHz).

Optical Microscopy and spectroscopy

Optical microscopy images were acquired using a Motic BA310 Me-T microscope (Motic, Barcelona, Spain) (equipped with a 50× 0.55 NA objective and an AMScope MU1803 CMOS Camera) and reflection spectra were collected from a spot of ~1.5–2 μm diameter with a Thorlabs CCS200/M fiber-coupled spectrometer (Thorlabs Inc., Newton, New Jersey, United States). More details about the micro-reflectance setup can be found in Reference [44].

3. Results and Discussion

Figure 1(a) shows the atomic force microscopy (AFM) topography of eight MoO_3 flakes with different thicknesses and Figure 1(b) shows their corresponding optical microscopy images. The direct comparison between the AFM and optical images allows us to build up a color-chart correlating the apparent color of the MoO_3 flakes deposited on top of the 297 nm SiO_2/Si substrate (the SiO_2 thickness is experimentally determined by reflectometry with ± 0.5 nm uncertainty) with their corresponding thickness with an uncertainty of ± 3 nm. Similar approaches have been reported for graphene [45], transition metal dichalcogenides [46], TiS_3 [47] and franckeite [48]. This method has the main limitation that it requires the use of a specific SiO_2 thickness as the interference colors of the MoO_3 flakes strongly depend on the underlying substrate. In this work we provide color-charts for four different nominal SiO_2 capping layer thicknesses: 297 nm, 271 nm, 148 nm and 88 nm. Figure 2 shows the correlation of optical images and AFM images for MoO_3 flakes transferred onto an 88 nm SiO_2/Si substrate. We address the reader to the Supporting Information for the data corresponding to the 271 nm and 148 nm thick SiO_2 substrates. Figure 3 shows a comparison between the apparent color vs. the thickness color-charts obtained for the four different SiO_2 thicknesses studied here. If a different SiO_2 thickness is wanted to be used, a calibration measurement like that shown in Figure 1 has to be carried out again for the desired SiO_2 thickness.

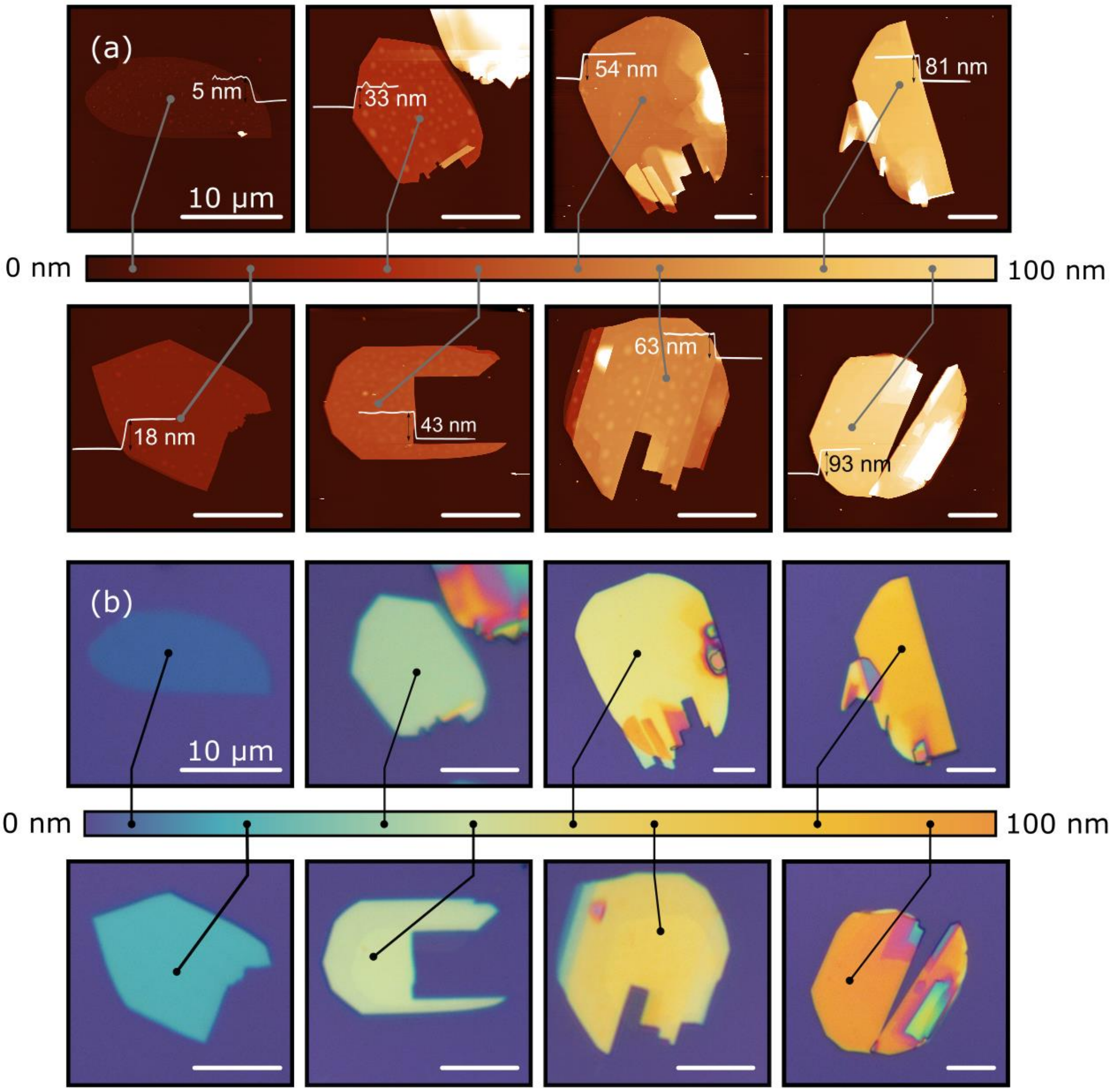

Figure 1. Thickness dependent apparent color of MoO_3 flakes. (a) AFM measurements of the exfoliated flakes with different thickness placed on a 297 nm SiO_2/Si substrate. **(b)** Optical images of the flakes and a colorbar with the apparent color of flakes with thickness from 5 nm up to 93 nm (from 7 to ~130 layers). Scale bars: 10 μm.

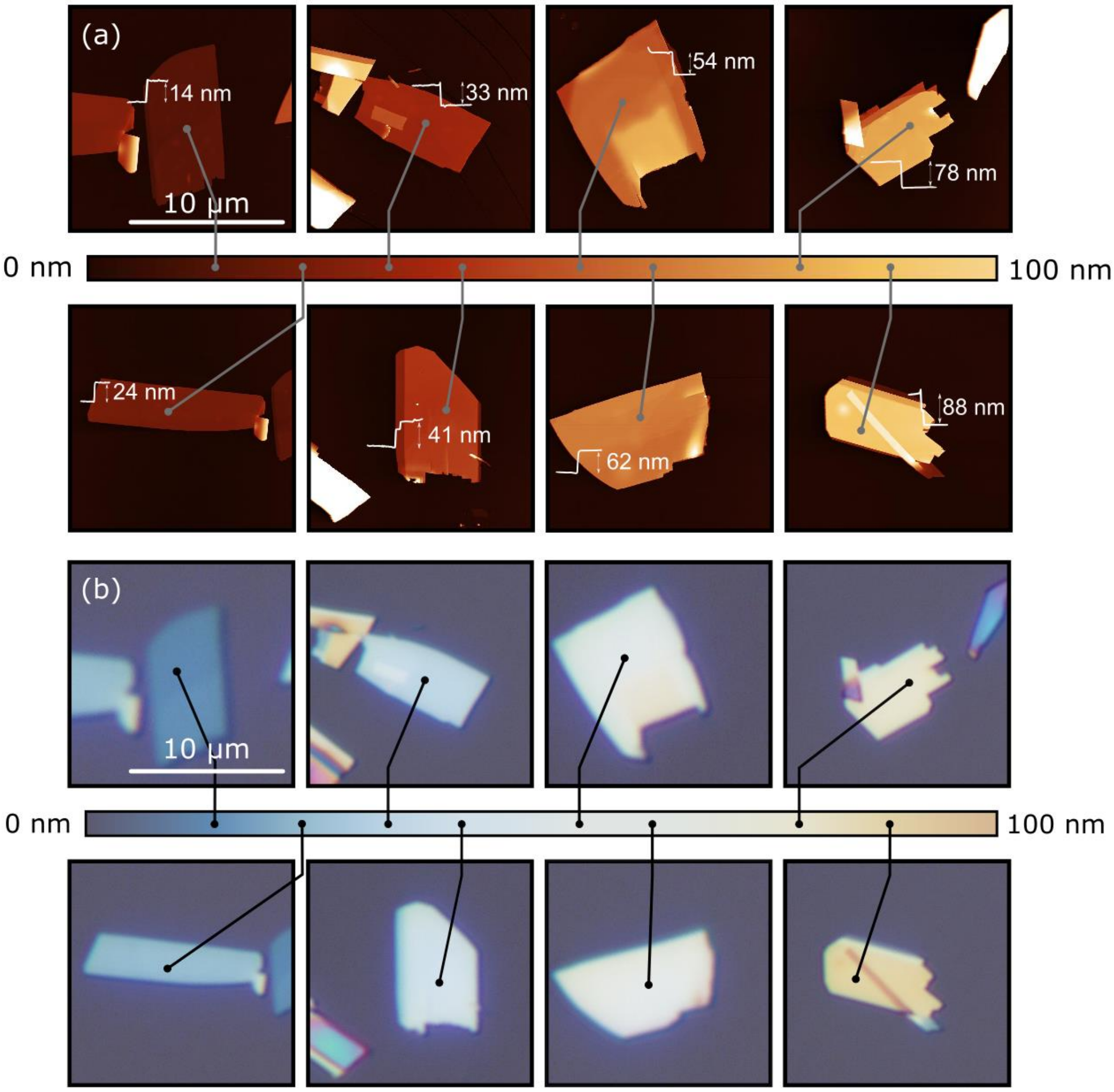

Figure 2. Thickness dependent apparent color of MoO_3 flakes. (a) AFM measurements of the exfoliated flakes with different thickness placed on an 88 nm SiO_2/Si substrate. **(b)** Optical images of the flakes and a colorbar with the apparent color of flakes with thickness up to 88 nm. Scale bars: 10 μm.

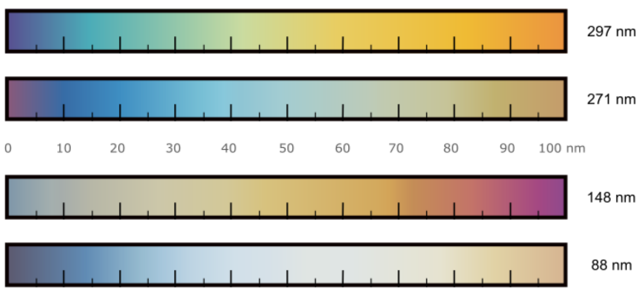

Figure 3. Thickness dependent apparent color of MoO_3 flakes on SiO_2/Si substrates with different oxide capping layer thickness.

Further, we quantitatively analyze the reflection spectra of MoO_3 flake to determine their thickness more accurately, similarly to previous works in transition metal dichalcogenides, muscovite mica and black phosphorus[44,49–51]. Differential reflectance spectra are acquired with a modified metallurgical microscope (BA 310 MET-T, Motic), details in Ref.[44]. A spectrum is first acquired onto the bare substrate (I_s) and then onto the flake (I_f) and the optical contrast (C) can be calculated as:

$$C = \frac{I_f - I_s}{I_f + I_s} \tag{1}$$

In Figure 4(a) we sketch the optical system, indicating the different optical media, used to model the optical contrast spectra. The refractive index values of the different media used for the model are shown in Figure S3(a). Figure 4(b) shows the experimental optical contrast obtained for different values of thickness of MoO_3.

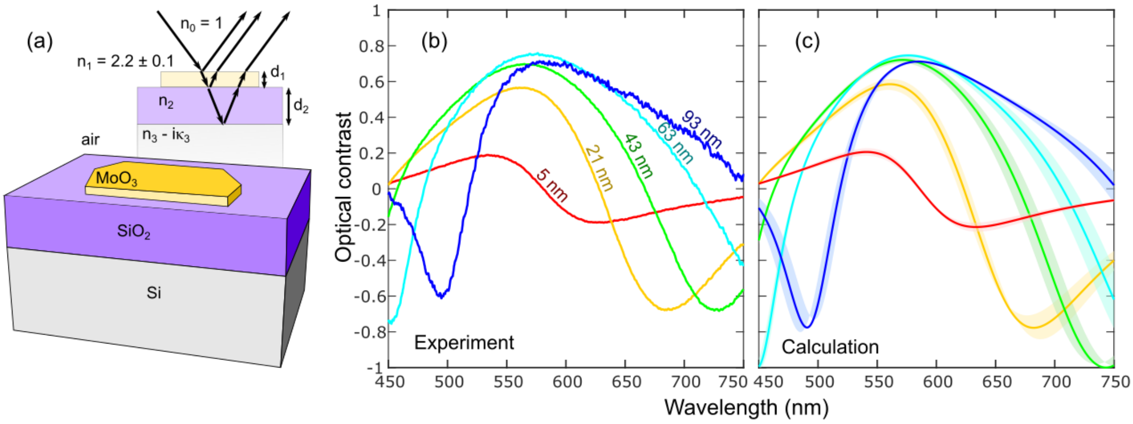

Figure 4. (a) Optical model used to calculate the MoO_3 optical contrast. (b) Optical contrast spectra measured on MoO_3 flakes, on a 297 nm SiO_2/Si substrate, of different thickness. (c) Calculated optical contrast (solid lines) using the Fresnel law-based model [52]. The shaded area accounts for an uncertainty of ±1 nm in the thickness of the flake [53].

The optical contrast of this kind of multilayer optical system can be calculated with high accuracy using a Fresnel law based model [54]. The reflection coefficient in a 4 media Fresnel model, is expressed as[55]:

$$r_4 = \frac{r_{01}e^{i(\Phi_1+\Phi_2)} + r_{12}e^{-i(\Phi_1-\Phi_2)} + r_{23}e^{-i(\Phi_1+\Phi_2)} + r_{01}r_{12}r_{23}e^{i(\Phi_1-\Phi_2)}}{e^{i(\Phi_1+\Phi_2)} + r_{01}r_{12}e^{-i(\Phi_1-\Phi_2)} + r_{01}r_{23}e^{-i(\Phi_1+\Phi_2)} + r_{12}r_{23}e^{i(\Phi_1-\Phi_2)}} \tag{2}$$

where sub index 0 refers to air, 1 to MoO_3, 2 to SiO_2 and 3 to Si. Assuming normal incidence, $\Phi_i = 2\pi\tilde{n}_i d_i/\lambda$ is the phase shift induced by the propagation of the light beam in the media i, being $\tilde{n}_i$, d_i and λ the complex refractive index, thickness of the media and wavelength, respectively; $r_{ij} = (\tilde{n}_i-\tilde{n}_j)/(\tilde{n}_i+\tilde{n}_j)$ is the Fresnel coefficient at the interface between the media i and j.

The reflection coefficient in a three media Fresnel model (*i.e.* the case of the bare substrate without MoO_3 flake), is expressed as:

$$r_3 = \frac{r_{01} + r_{12}e^{-i2\Phi_1}}{1 + r_{01}r_{12}e^{-i2\Phi_1}} \tag{3}$$

Where sub index 0 is air, 1 is SiO_2 and 2 is Si. With these equations, one can calculate the optical contrast, firstly calculating the reflected intensity:

$$R_k = |\overline{r_k} r_k|, \forall\, k = 3, 4 \tag{4}$$

Then the optical contrast can be defined through the following operation that correlates the reflected intensity by the bare substrate (R_3) with the reflected intensity by the MoO_3 flake (R_4) as:

$$C = \frac{R_4 - R_3}{R_4 + R_3} \tag{5}$$

Interestingly, we found that one can accurately reproduce the experimental optical contrast spectra by simply assuming a refractive index of MoO_3 of $\tilde{n}_1$ = 2.2 – i0. Note that, according to Reference [35], the band structure of MoO_3 has a negligible thickness dependence and thus we do not expect the refractive index to depend on the flake thickness. The results of the calculated spectra with this model are depicted in Figure 4(c). The real part of the refractive index of bulk MoO_3 n_{MoO_3} spans in the literature from 1.8 to 2.3 [14,56,57]. Since the literature values of the extinction coefficient in the visible range are very low, we neglect that term in our calculations, still providing good fit to the experimental data. As a verification Figure S3(b) shows the complex refractive index as a function of wavelength, $\tilde{n}=n+i\kappa$, measured with ellipsometry, of polycrystalline MoO_3 films, showing values of n = 2-2.3 in the visible part of the spectrum (from 1.5 to 3 eV), and a negligible value of κ in the same region (from 0.02 to 0.1).

Some of the features present in the optical contrast spectra, like the local maxima and minima, strongly depend on the thickness of the MoO_3 medium. n fact, the shape of the optical contrast spectra, including the maxima and minima features, arises from the interference colors effect. It is therefore clear that these features will depend on the thickness of the MoO_3 flake as the optical paths of the light beams passing through flakes with different thicknesses will be different. Therefore, one can evaluate the thickness of a MoO_3 flake by calculating the optical contrast according to Equations (2)–(5) for different thicknesses of MoO_3 and determining the best fit by minimum squares.

Figure 5(a) compares the optical contrast measured for a MoO_3 flake (16.5 nm thick according to the AFM) with the optical contrast calculated assuming a thickness in the range of 0-100 nm. The best matching is obtained for a thickness of 13.5 nm. The inset in Figure 5 (a) shows the square of the difference between the measured contrast and the calculated one as a function of the thickness assumed for the calculation. The plot shows a well-defined minimum at thickness 13.5 nm.

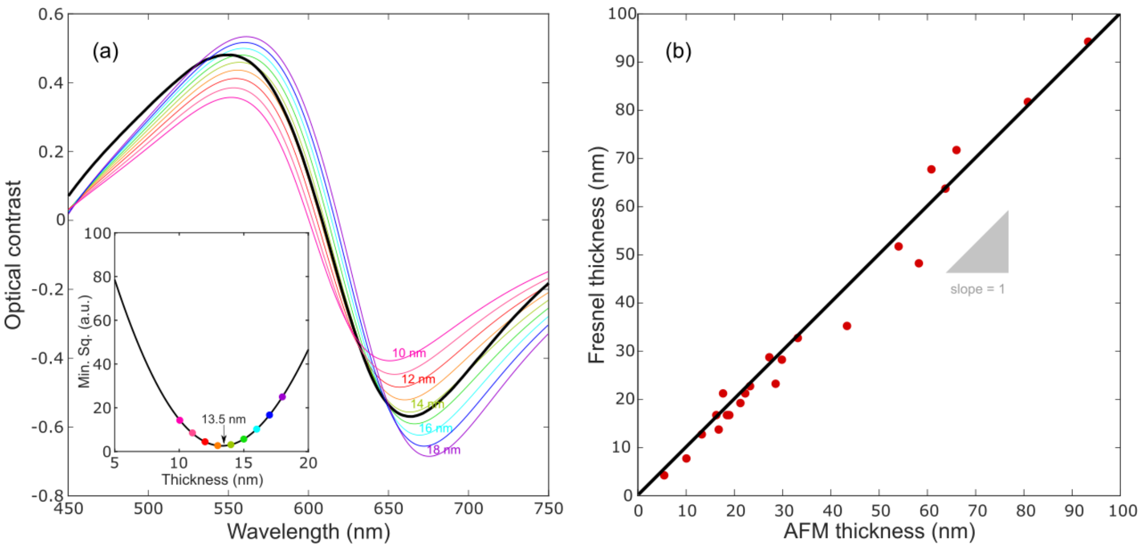

Figure 5. (a) Experimental optical contrast (black) and comparison with the calculated ones (color lines) of MoO_3 flakes, with thicknesses from 10 to 18 nm, placed onto a substrate of 297 nm of SiO_2. Inset of figure: minimum square value for different values of thickness of the flakes. **(b)** Comparison of thickness measured with AFM and the

Fresnel model. Experimental data represented as red dots and line in blue has a slope of 1.

In order to benchmark this thickness determination method, Figure 5 (b) compares the thickness values measured with AFM for 23 flakes from 5 nm to 100 nm thick, with the values obtained following the optical contrast fit method discussed above. In the plot we include a straight line with a slope of 1 that indicates the perfect agreement between the thickness of MoO_3 nanosheets determined by AFM and the fit to the Fresnel law based model. The low dispersion along the slope = 1 line indicates the good agreement between the thickness determined by both methods. In effect, the calculated linear regression of the data points in the Figure 5 (b) has a slope of 1.02.

4. Conclusions

In summary, we provided two fast and non-destructive complementary methods to evaluate the number of layers of MoO_3 nanosheets using optical microscopy. First, one can get a coarse estimation of the thickness (with ±3 nm of uncertainty) by comparing the apparent color of the flakes with a pre-calibrated color-chart. This method is very fast, but it requires the previous calibration of a color-chart that depends on the SiO_2 capping layer thickness (nonetheless, we provide pre-calibrated color-charts for four commonly used SiO_2 capping layer thicknesses). The second method is based on the measurement of the optical contrast spectrum of the MoO_3 flake under study and the subsequent fit to a Fresnel law-based model that includes the optical constants of each medium. This method requires a modification of the optical microscope to allow for differential reflectance measurements, but it can provide lower uncertainty (±2 nm of median standard deviation for the samples on 297 nm of SiO_2), and it can be easily employed for flakes transferred on SiO_2 /Si substrates with different SiO_2 thicknesses. We believe that the development of these thickness determination methods can be very helpful for the community working on MoO_3 as it can effectively speed up the research on this 2D material.

Author Contributions: SP has fabricated the MoO_3 flake samples and performed the experimental work. AM-J, RSG, have performed the ellipsometry measurements on the polycrystalline MoO_3 film. CM and AC-G have supervised and designed the experiments. SP and AC-G have drafted the firsts version of the manuscript. All authors have contributed to write the final version of the manuscript.

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Conflicts of Interest: The authors declare no conflict of interest.

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Supplementary Materials:

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In order to provide more information about the color of MoO_3 flakes deposited on SiO_2/Si substrates, in Figures S1 and S2 are depicted the color-chart that corresponds to 148 nm and 271 nm of SiO_2, respectively. The thicknesses of these flakes (shown at the corner of each image) are calculated using the equations (2)-(5) and are expected to be accurate with approximately ± 1.6 nm of uncertainty, as we have calculated in the previous SiO_2/Si substrates in Figures 1 and 2.

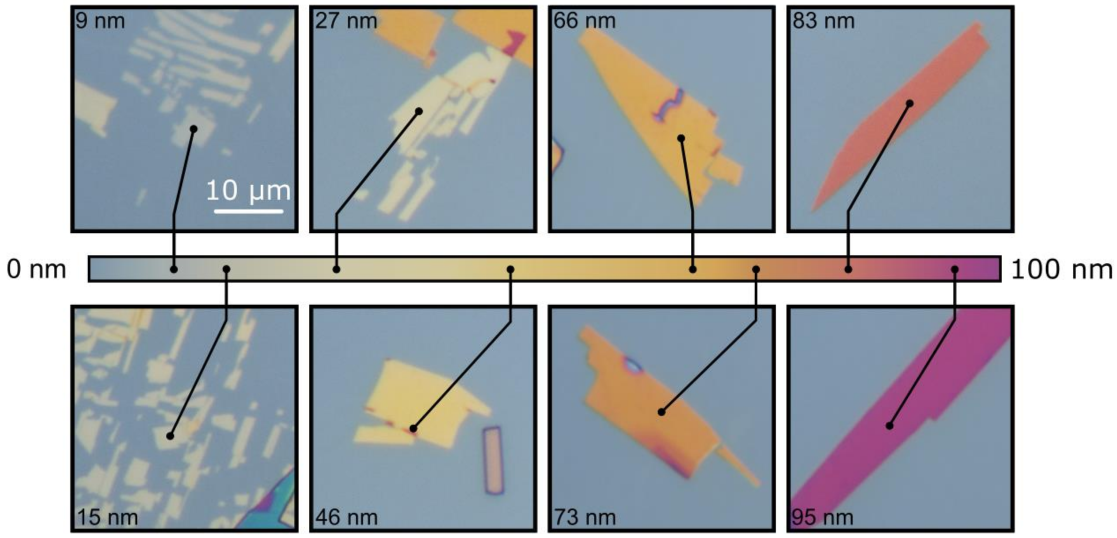

Figure S1: Color-chart of MoO_3 on SiO_2/Si with 148 nm of SiO_2 capping layer.

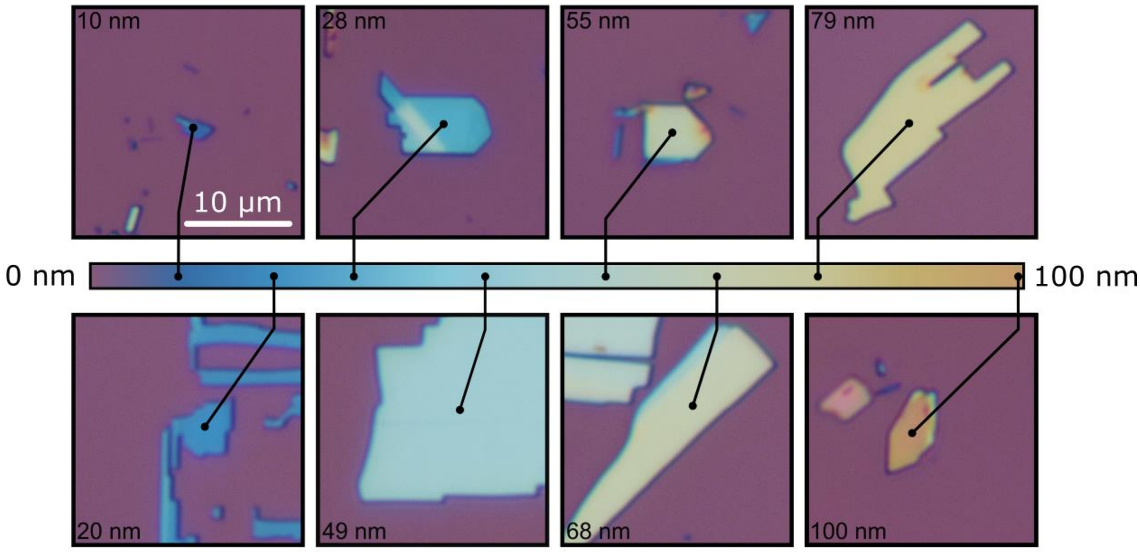

Figure S2: Color-chart of MoO_3 on SiO_2/Si with 271 nm of SiO_2 capping layer.

We have determined, from spectroscopic ellipsometry measurements, the optical constants of continuous α-MoO_3 films grown onto SiO_2/Si substrates with an oxide thickness of 280 nm. For the determination of the complex refractive ñ=n+iκ we have assumed a 4 media model, and we have considered the dispersion values of the refractive index for both Si and SiO_2. The resulting components n and κ as a function of the wavelength are depicted in Figure S3 (b). They have been fitted in the full UV-NIR range (275-1700nm) with a general oscillator model that is a sum of a Tauc-Lorentz oscillator with a bandgap at 3.4 eV, and a Lorentz oscillator for the UV region. Here we show the resulting visible part of the spectrum. It is shown that, in this part of the spectrum, the absorption is negligible ($\kappa < 0.085$), according with the excellent transparency of the flakes in the visible region, and thus it is reasonable to approximate $\kappa = 0$.

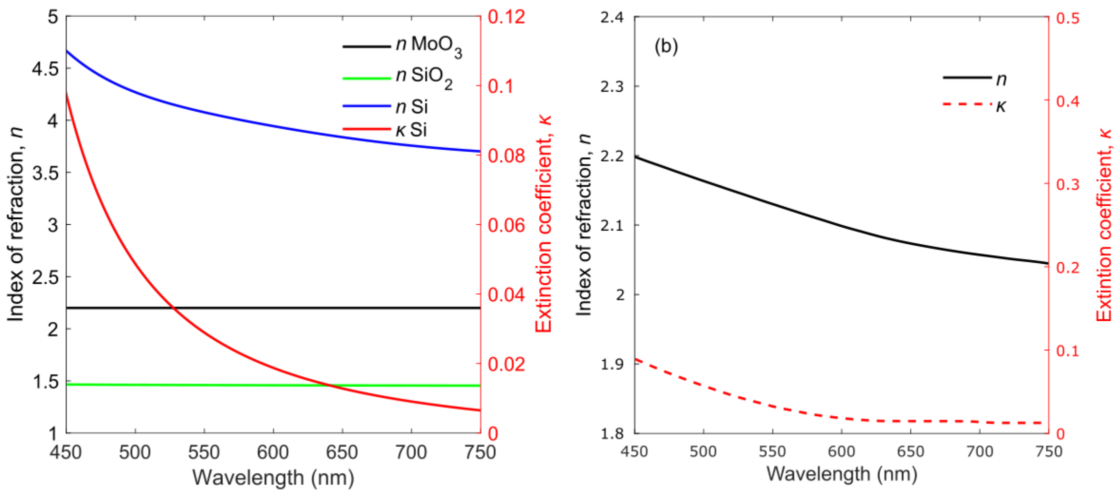

Figure S3: (a) Index of refraction of MoO_3 films, SiO_2 and Si materials. These values are obtained from bibliography [57–60] and they are in accordance with our spectroscopic ellipsometry measurements. **(b)** Index of refraction (n) and extinction coefficient (κ) measured by ellipsometry on a pollycrystalline α-MoO_3 film transferred onto a SiO_2/Si substrate with 280 nm of SiO_2 thickness layer.

Using equations (2)–(5), one can calculate the dependence of the optical contrast for a monolayer MoO_3 flake as a function of the illumination wavelength and the SiO_2 substrate. The results of those calculations are shown in Figure S4. This figure gives the chance to determine the substrate that optimizes its optical identification. For molybdenum trioxide, the SiO_2 thickness values that enhances the optical contrast at a wavelength of 550 nm (where the performance of the human eye is better [61]) are 70, 110, 260 and 300 nm, where the optical contrast is maximum.

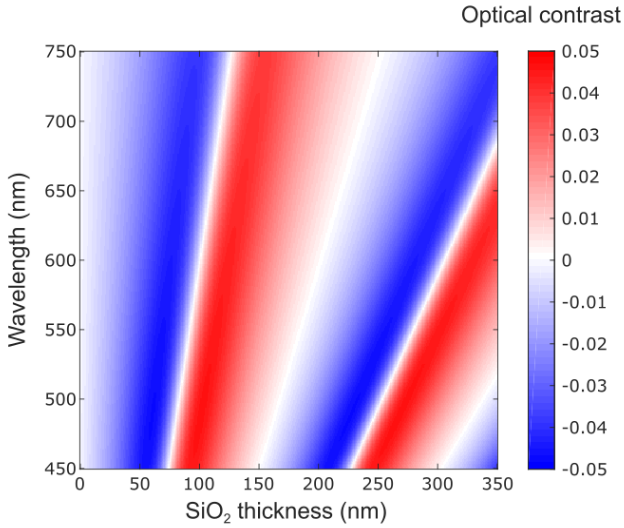

Figure S4: Calculated optical contrast dependence on illumination wavelength and SiO_2 thickness for a monolayer MoO_3 on a SiO_2/Si substrate.